

CC-SPR: Geometry-Sensitive Sparse Persistent Representatives for Interpretable Omics Feature Selection

A Cross-Domain Evaluation Across Cancer and Plant Transcriptomics

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Abstract

Persistent homology provides a multiscale topological summary of omics data, but standard pipelines discard the geometric identity of detected cycles and rely on fixed Euclidean metrics. We introduce CC-SPR (Cycle-Consistent Sparse Persistent Representatives), which addresses both limitations within the harmonic persistent homology scaffold. First, we replace the dense ℓ_2 representative with a convex elastic-net objective that concentrates cycle support on a small number of edges, yielding interpretable, bootstrap-stable feature-importance profiles. Second, we formalize the sample-space metric as a selectable *backend*, enabling systematic comparison of Euclidean, Ollivier–Ricci, diffusion, PHATE-like, and distance-to-measure (DTM) geometries within a unified filtration–representation–projection framework. All five backends are benchmarked on three cancer-genomics datasets and a plant developmental atlas. Backend ranking is dataset-dependent and scale-sensitive: DTM leads on pilot LUAD (F1=0.5315); Ricci leads at full scale (F1=0.6339); Euclidean leads on GSE161711 (F1=0.7483). DTM uniquely recovers xenobiotic metabolism via cytochrome P450 on LUAD (KEGG; adj. $p=7\times 10^{-4}$), and zero genes are shared across all six backend configurations. CC-SPR produces $2.8\times$ sparser sample-bar matrices than harmonic representatives while preserving comparable clustering structure. Source code: https://github.com/ChimdiWalter/Omics_Project_CCSPR.

Keywords: topological data analysis, persistent homology, omics feature selection, geometry-aware, sparse optimization

1 Introduction

1.1 Biological motivation

Modern omics experiments—bulk RNA-seq, single-cell transcriptomics, methylation arrays, proteomics—routinely produce data matrices with n samples in 10^1 – 10^5 and p features in 10^3 – 10^5 . These matrices encode geometry that reflects biological structure: tumor subtypes form clusters; differentiation trajectories trace manifold-like paths; regulatory circuits create loop-like patterns in expression space; and microenvironment heterogeneity introduces curvature and density variation. This geometric richness makes omics a natural setting for topological data analysis (TDA) (Carlsson, 2009; Edelsbrunner and Harer, 2010), which has been applied to gene expression landscapes (Rizvi et al., 2017), protein structure (Xia and Wei, 2014), tumor microenvironments (Vipond et al., 2021), single-cell trajectories (Moon et al., 2019), neural population coding (Giusti et al., 2016), and viral evolution (Chan et al., 2013).

Persistent homology, the workhorse of TDA, captures collective structure—loops, voids, connected components—that cannot be detected by feature-wise statistics or pairwise correlations alone. However, the standard pipeline has a well-known interpretability gap: while a barcode tells us *that* a topological feature exists and *how long* it persists, it does not tell us *which samples* participate in the corresponding cycle, nor *which molecular features* drive it.

1.2 The harmonic PH advance and its limitations

Harmonic persistent homology (Gurnari et al., 2025) addresses this gap by computing a representative cycle for each persistent class that minimizes an ℓ_2 (harmonic) energy, creating a bridge from topological features back to individual samples and molecular features. However, two practical limitations motivate the present work:

1. **Dense representatives blur interpretation.** The harmonic ℓ_2 objective spreads weight broadly across edge support. In omics, where the goal is to identify compact gene sets associated with topological signals, dense support complicates downstream pathway analysis.
2. **Euclidean geometry may be inappropriate.** The default sample-space metric is Euclidean distance, which is often too rigid for biological data with nonuniform density, curvature-rich manifold structure, or developmental trajectories. Different metrics can produce different filtrations and different representatives—a sensitivity that should be modeled explicitly.

1.3 Contributions

We introduce **CC-SPR** (Cycle-Consistent Sparse Persistent Representatives), which addresses both limitations within the harmonic persistent homology scaffold:

1. **Sparse elastic representatives.** We replace the dense ℓ_2 objective with a convex elastic-net objective that concentrates cycle support on a small number of edges, yielding interpretable, bootstrap-stable feature-importance profiles.
2. **Backend-aware geometry.** We formalize the sample-space metric as a selectable backend, implementing Euclidean, Ollivier–Ricci, diffusion, PHATE-like, and distance-to-measure (DTM) geometry families.
3. **Topological influence profiles (TIP).** We introduce a bootstrap-aggregated feature-stability diagnostic that enables direct comparison of support concentration across geometries.
4. **Cross-domain evaluation.** We benchmark all five backends on three cancer-genomics datasets and provide targeted cross-domain evidence on plant transcriptomics.

CC-SPR is not a claim of predictive superiority over standard classifiers. It is a framework that makes topology-derived features more interpretable and geometry-aware; every harmonic PH analysis is a special case with $\lambda_1 = 0$ and a Euclidean backend.

2 Materials and Methods

2.1 Datasets

We use four datasets with distinct evaluation roles (Table 1):

- **TCGA-LUAD** (lung adenocarcinoma, $p \approx 20,000$): main benchmark anchor. Evaluated at two scales: pilot ($n = 60$, 2-split CV, $n_{\text{boot}} = 5$) and full cohort ($n = 275$, 5-fold CV, $n_{\text{boot}} = 5$, 2000 variable genes).
- **GSE161711** (CLL/venetoclax bulk RNA-seq, $n = 96$): disease-aligned bulk validation with 5-fold CV.
- **GSE165087** (CLL/RS-like scRNA-seq, bounded $n = 320$): bounded single-cell feasibility test with all five backends.

- **Arabidopsis root atlas** (GSE123013, [Denyer et al. 2019](#), scRNA-seq, $n = 500$, 16 Leiden clusters): targeted cross-domain generalization test with Euclidean and DTM only.

Dataset	Type	n	p	Labels
TCGA-LUAD (pilot)	Bulk RNA-seq	60	20,530	Molecular subtype
TCGA-LUAD (full)	Bulk RNA-seq	275	20,530	Molecular subtype
GSE161711	Bulk RNA-seq	96	17,860	Venetoclax response
GSE165087	scRNA-seq	320	5,000	CLL vs Richter
Arabidopsis root	scRNA-seq	500	21,965	Leiden cluster (16 types)

Table 1: Datasets used in this study. GSE161711 and GSE165087 are public GEO substitutes for originally intended controlled-access datasets.

2.2 The CC-SPR framework

CC-SPR preserves the four-step harmonic persistent homology scaffold—construct a sample-space metric, compute a filtration and persistent homology, choose a representative cycle, project to feature space—but modifies two steps: the representative objective (Step 3) and the metric (Step 1).

Notation. Let $X \in \mathbb{R}^{n \times p}$ denote an omics data matrix. A *geometry backend* $\mathcal{G}$ with hyperparameters $\theta_{\mathcal{G}}$ induces a symmetric distance matrix $D^{(\mathcal{G})} \in \mathbb{R}_{\geq 0}^{n \times n}$. The Vietoris–Rips filtration on $D^{(\mathcal{G})}$ produces a nested sequence of simplicial complexes whose homology yields a persistence barcode $\{[b_i, d_i)\}$ ([Zomorodian and Carlsson, 2005](#); [Edelsbrunner and Harer, 2010](#)). We focus on H_1 bars because 1-cycles in sample space project to feature space via the data matrix, creating a bridge from topological structure to gene-level influence.

Harmonic (dense) objective. For a selected H_1 bar $[b, d)$ with initial cycle $z_0 \in C_1$ and boundary operator ∂_2 , the harmonic representative minimizes

$$\hat{y}_{\text{harm}} = \arg \min_{y \in C_2} \|z_0 + \partial_2 y\|_2^2. \quad (1)$$

This produces the smoothest cycle, typically with dense edge support ($\|z\|_0 = O(|E|)$), which dilutes biological signal when projected to feature space ([Gurnari et al., 2025](#); [Friedman, 1998](#)).

Sparse elastic objective. CC-SPR replaces (1) with an elastic-net formulation:

$$\hat{y}_{\text{spr}} = \arg \min_{y \in C_2} \underbrace{\|z_0 + \partial_2 y\|_2^2}_{\text{fidelity}} + \underbrace{\lambda_1 \|z_0 + \partial_2 y\|_1}_{\text{sparsity}} + \underbrace{\lambda_2 \|y\|_2^2}_{\text{stability}}, \quad (2)$$

with $\lambda_1, \lambda_2 \geq 0$. The objective is convex (strongly convex when $\lambda_2 > 0$), and the ℓ_1 penalty drives small edge weights to zero, producing representatives with compact support (see Supplementary Section S2 for proofs of convexity and sparsity guarantees). The structure mirrors the elastic net (Zou and Hastie, 2005): λ_1 plays the LASSO role promoting sparsity, λ_2 provides ridge-like stability, and the “design matrix” is the boundary operator ∂_2 .

2.3 Topological influence profiles (TIP)

For bootstrap replicates $b = 1, \dots, B$ and a top- K feature selection threshold, let S_b denote the set of features with the K largest absolute projected weights from the sparse representative in replicate b . The *topological influence profile* is

$$\text{TIP}(j) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{j \in S_b\}, \quad j = 1, \dots, p.$$

Concentrated TIP mass (a few features with $\text{TIP}(j) \approx 1$) indicates stable topological signal; diffuse mass indicates unstable support. Because TIP applies identically to any geometry backend, it provides a natural diagnostic for comparing how backends affect interpretability.

2.4 Geometry backends

A central design principle of CC-SPR is that the sample-space metric is a selectable parameter, not a fixed assumption. Backend choice can be understood as a controlled perturbation: when the distance matrices from two backends differ by ε_∞ in the ℓ_∞ norm, the bottleneck distance between persistence diagrams is bounded by ε_∞ (Chazal et al., 2016). We implement five backends:

Euclidean (baseline). Standard ℓ_2 distance in PCA space: $D_{ij}^{(\text{eu})} = \|x_i - x_j\|_2$. Simple and efficient, but rigid—it treats all directions equally and does not adapt to local density or curvature.

Ollivier–Ricci curvature. The Ollivier–Ricci curvature (Ollivier, 2007, 2009) of an edge (u, v) is $\kappa(u, v) = 1 - W_1(\mu_u, \mu_v)/d(u, v)$, where W_1 is the Wasserstein-1 distance between

neighborhood measures. Discrete Ricci flow (Ni et al., 2019) iteratively updates edge weights: $w_{uv}^{(t+1)} = w_{uv}^{(t)} - \eta \cdot \kappa^{(t)}(u, v) \cdot w_{uv}^{(t)}$, upweighting negatively curved (bridge) edges. This delays merger of distinct components in the filtration, potentially preserving genuine cycles at the expense of noise-induced shortcuts.

Diffusion. Diffusion distances (Coifman and Lafon, 2006) are derived from the eigendecomposition of a Markov transition matrix $P = D_W^{-1}W$, capturing connectivity that persists across random walks of varying length. A time parameter t controls scale: small t recovers local geometry; large t emphasizes global connectivity. Implemented via truncated eigendecomposition (50 eigenvectors).

PHATE-like. PHATE (Moon et al., 2019) transforms diffusion operators via a log potential: $U_i = -\log(P_{i,\cdot}^t + \epsilon)$, with potential distance $D^{(\text{phate})}(i, j) = \|U_i - U_j\|_2$. This measures how differently random walks from i and j explore the graph, approximating a Fisher–Rao-like distance. Our implementation uses a lightweight potential-distance computation without the full MDS step.

Distance-to-measure (DTM). DTM (Chazal et al., 2011, 2017) replaces pairwise distances with density-smoothed variants: $d_{\mu_X, m_0}(x) = (\frac{1}{|m_0 n|} \sum_{j=1}^{\lceil m_0 n \rceil} \|x - x_{(j)}\|^2)^{1/2}$. DTM-weighted distances delay inclusion of outlier-adjacent simplices, suppressing topological noise from density spikes—relevant for omics data with batch effects, dropout, or technical outliers.

2.5 Evaluation protocol

All five backends are evaluated under the same pipeline: backend $\rightarrow$ Rips filtration $\rightarrow$ persistent homology $\rightarrow$ sparse representative $\rightarrow$ TIP $\rightarrow$ feature selection $\rightarrow$ weighted-F1 classification (logistic regression). Dataset-appropriate replication:

- LUAD pilot: 2-split CV, $n_{\text{boot}} = 5$; all five backends.
- LUAD full: 5-fold CV, $n_{\text{boot}} = 5$, 2000 variable genes; all five backends.
- GSE161711: 5-fold CV, $n_{\text{boot}} = 5$; all five backends.
- GSE165087: 1 split, $n_{\text{boot}} = 1$; all five backends.
- Arabidopsis: 3-fold CV, $n_{\text{boot}} = 3$; Euclidean and DTM only.

Leakage control. Within each training fold: (1) variable genes are selected by variance on training data only; (2) standardization uses training-derived parameters; (3) bootstrap TIP computation uses training-fold subsamples only; (4) feature selection and classifier training operate exclusively on training data. Two departures from a fully strict within-fold protocol: global PCA where applied (single-cell data), and full-dataset variable-gene selection in the standard baseline only at full LUAD scale. Both are documented and apply identically across all methods.

Resource policy. Experiments were conducted in two phases: Phase 1 on an HPC cluster (48 GB, 8 CPUs; pilot LUAD, GSE161711, GSE165087) and Phase 2 on a local workstation (Intel i9-9920X, 24 threads, 125 GB RAM; full-scale LUAD and Arabidopsis). Rather than silently omitting failed runs, completed outputs are reused and resource limits are stated explicitly.

2.6 Pipeline algorithm

Algorithm 1 CC-SPR Pipeline

Require: Data matrix $X \in \mathbb{R}^{n \times p}$, geometry backend $\mathcal{G}$, parameters $\lambda_1, \lambda_2, k, K$, bootstraps B

Ensure: Topological influence profile $\text{TIP} \in [0, 1]^p$

- 1: Compute distance matrix $D^{(\mathcal{G})} \leftarrow \text{Backend}(X, \mathcal{G})$
 - 2: Compute Vietoris–Rips filtration and persistent homology on $D^{(\mathcal{G})}$
 - 3: Select the most persistent H_1 bar $[b, d]$
 - 4: Extract initial cycle z_0 and boundary operator ∂_2 at scale $\varepsilon = (b + d)/2$
 - 5: **for** $b = 1, \dots, B$ **do**
 - 6: Draw bootstrap resample $X^{(b)}$ of size n
 - 7: Recompute $D^{(\mathcal{G})^{(b)}}$, filtration, and persistent homology
 - 8: Solve (2) to obtain $\hat{z}_{\text{spr}}^{(b)}$
 - 9: Project $\hat{z}_{\text{spr}}^{(b)}$ to feature space: $w^{(b)} \in \mathbb{R}^p$
 - 10: Select top- K features: $S_b \leftarrow \{j : |w_j^{(b)}| \text{ among top-}K\}$
 - 11: **end for**
 - 12: Compute $\text{TIP}(j) \leftarrow \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{j \in S_b\}$ for each feature j
 - 13: **return** TIP
-

Computational complexity. The dominant costs are distance matrix computation ($O(n^2p)$ for Euclidean; higher for Ricci flow and diffusion eigendecomposition), persistent homology ($O(m^3)$ worst case, much faster in practice; [GUDHI Project 2014](#)), and sparse optimization ($O(|E|^2 \cdot |F|)$ per bootstrap). Total cost is $O(B \cdot (n^2p + m^3))$, dominated by the

bootstrap loop. Full-scale LUAD ($n = 275$) requires 35 minutes to 2.5 hours per backend;
the CVXPY elastic-net solver on the boundary matrix is the computational bottleneck.

2.7 Comparison with harmonic PH

Table 2 provides a systematic comparison of harmonic PH and CC-SPR.

Axis		Harmonic PH	CC-SPR
Metric assumption	as-	Fixed Euclidean	Selectable back-end family
Representative criterion		Dense ℓ_2 (harmonic)	Sparse elastic net
Sparsity		No explicit control	λ_1 -controlled
Interpretability		Diffuse feature weights	Concentrated feature weights
Stability diagnostics	diag-	Not built-in	TIP profiles (bootstrap)
Computational cost		Moderate	Moderate to high (backend-dependent)
Backend flexibility	flexi-	None (single metric)	Euclidean, Ricci, diffusion, PHATE, DTM
Biological alignment		Medium (diluted signal)	High (compact gene sets)

Table 2: Comparison of harmonic PH and CC-SPR. CC-SPR extends harmonic PH by adding sparsity control, stability diagnostics, and geometry awareness while preserving the same scaffold.

The harmonic representative is sufficient when cycle support is naturally compact, downstream tasks do not require feature attribution, or datasets are small enough that dense support does not dilute signal. CC-SPR adds value when compact gene-set identification, geometry sensitivity analysis, or bootstrap stability assessment is required.

3 Results

3.1 Backend ranking is dataset-dependent

The central empirical question is whether the five geometry backends produce meaningfully different results across datasets. Table 3 presents the complete geometry-family benchmark.

Dataset	Backend	Mean F1	Std F1	N splits
LUAD ($n = 60$)	euclidean	0.4342	0.0394	2
LUAD	ricci	0.5012	0.1073	2
LUAD	diffusion	0.5000	0.1000	2
LUAD	phate_like	0.3546	0.0880	2
LUAD	dtm	0.5315	0.0765	2
GSE161711 ($n = 96$)	euclidean	0.7483	0.0825	5
GSE161711	ricci	0.5944	0.0845	5
GSE161711	diffusion	0.5553	0.0721	5
GSE161711	phate_like	0.5396	0.0632	5
GSE161711	dtm	0.6178	0.1333	5
GSE165087 (bounded, $n = 320$)	all backends	0.9254	0.0	1
Arabidopsis root ($n = 500$)	euclidean	0.5933	0.1660	3
Arabidopsis root	dtm	0.6664	0.0940	3
Arabidopsis root	standard	0.9585	0.0170	3
LUAD full ($n = 275$)	euclidean	0.5655	0.0361	5
LUAD full	ricci	0.6339	0.0503	5
LUAD full	diffusion	0.5522	0.0609	5
LUAD full	phate_like	0.5445	0.0640	5
LUAD full	dtm	0.5814	0.0611	5
LUAD full	standard	0.7501	0.0294	5

Table 3: Geometry-family benchmark across four datasets. Backend ranking is dataset-dependent and scale-sensitive: on pilot LUAD, DTM $>$ Ricci $\approx$ Diffusion $>$ Euclidean $>$ PHATE-like; at full scale, Ricci overtakes DTM (0.6339 vs 0.5814); on GSE161711, Euclidean $\gg$ all others.

On pilot LUAD, DTM leads (0.5315), consistent with density-aware distances better capturing heterogeneous subtype topology. At full scale ($n = 275$), Ricci overtakes DTM (0.6339 vs 0.5814), suggesting curvature correction benefits from denser point clouds. On GSE161711, Euclidean dominates (0.7483, +0.1305 over DTM), indicating that local PCA neighborhoods suffice for this more homogeneous cohort. All backends produce identical F1 on the resource-bounded GSE165087 configuration.

Takeaway: No single backend is universally best; backend ranking depends on dataset structure and sample size.

3.2 Comparison with harmonic and standard baselines

Table 4 provides historical context from the initial two-backend evaluation.

Expanding the backend family and increasing sample size both improve CC-SPR: the original Ricci-only result (F1 = 0.3133) improves to DTM = 0.5315 at pilot scale and Ricci = 0.6339 at full scale.

Dataset	Model	Mean F1	N
LUAD ($n = 60$)	standard	0.7670	2
LUAD ($n = 60$)	harmonic	0.4044	2
LUAD ($n = 60$)	ccspr (Ricci)	0.3133	2
GSE161711	standard	0.9711	5
GSE161711	harmonic	0.5850	5
GSE161711	ccspr (Ricci)	0.6028	5
GSE165087 (bounded)	all methods	0.9254	1

Table 4: Historical baseline comparison from the initial two-backend evaluation. Standard baselines dominate on raw F1; the gap narrows from 0.236 at pilot scale to 0.116 at full scale.

Takeaway: Standard baselines remain strongest on raw F1, but the CC-SPR–standard gap narrows with scale and backend optimization.

3.3 LUAD multiseed paired statistics

The multiseed analysis measures paired differences across 18 repeated splits, controlling for split-specific data variation (Table 5).

Comparison	n	Mean Δ	Paired t (p)	Wilcoxon (p)
CC-SPR vs harmonic	18	+0.027	0.514	0.495
CC-SPR vs standard	18	−0.186	2.7×10^{-4}	3.3×10^{-4}
Ricci sparse vs Eu. sparse	18	+0.046	0.308	0.265

Table 5: LUAD multiseed paired statistics ($n = 18$ splits). CC-SPR significantly underperforms the standard baseline ($p < 0.001$). The Ricci advantage (+0.046) is substantively interesting but not significant at this sample size.

The five-backend results corroborate the Ricci direction: Ricci exceeds Euclidean at both pilot (+0.067) and full scale (+0.068).

3.4 Topological diagnostics across the geometry family

Beyond classification, the five-backend evaluation provides rich topological diagnostics (Tables 6 and 7).

Key findings: Euclidean’s highest H_1 lifetime and TIP Gini on LUAD do not translate to the highest classification F1. On GSE161711, Ricci creates the most persistent topological features (H_1 lifetime = 0.1248) but trails Euclidean on classification, illustrating that persistent topology captures structure complementary to class separation. DTM’s pro-

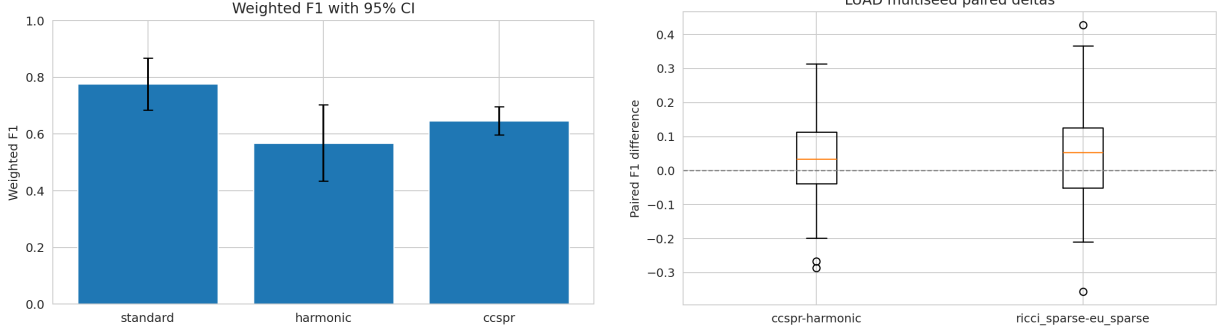


Figure 1: **Left:** LUAD benchmark showing the predictive gap between standard baseline and topology-derived models. **Right:** Paired multiseed deltas across 18 splits.

Backend	n_{ok}	$\bar{\ell}$	σ_{ℓ}	$\bar{\rho}$	σ_{ρ}	Support	TIP Ent.	TIP Gini
euclidean	5	0.0798	0.0250	0.0469	0.0296	50	3.7327	0.3212
ricci	4	0.0130	0.0067	0.0067	0.0080	71	4.2261	0.0982
diffusion	5	0.0256	0.0177	0.0146	0.0145	78	4.2740	0.1797
phate_like	4	0.0112	0.0100	0.0085	0.0076	80	4.3820	0.0000
dtm	5	0.0083	0.0059	0.0068	0.0068	76	4.2725	0.1642

Table 6: LUAD topological diagnostics. Euclidean produces the highest H_1 lifetime (0.0798) and most concentrated TIP (Gini = 0.3212). PHATE-like produces zero Gini, suggesting over-smoothing.

jected support on GSE161711 (24 features vs 59–95) demonstrates the concentration effect
of outlier-robust distances.

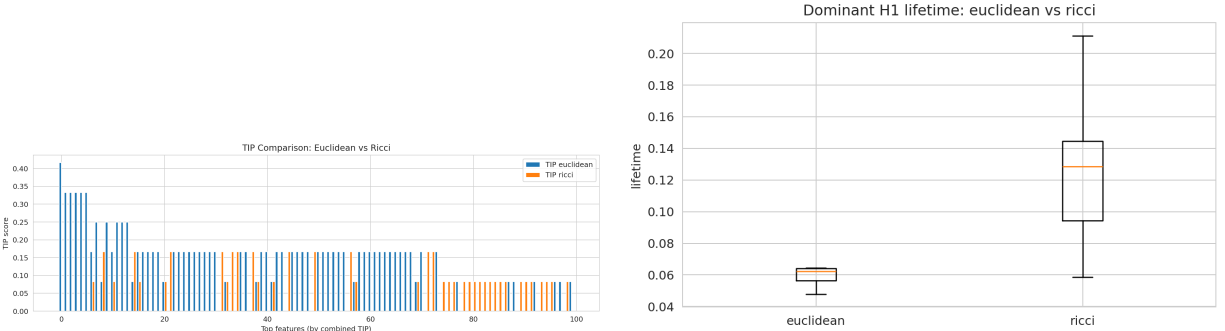


Figure 2: LUAD TIP and H_1 lifetime diagnostics comparing Euclidean and Ricci backends.

Takeaway: Topological diagnostics vary systematically across backends, providing in-
formation unavailable from classification metrics alone.

Backend	n_{ok}	$\bar{\ell}$	σ_{ℓ}	$\bar{\rho}$	σ_{ρ}	Support	TIP Ent.	TIP Gini
euclidean	5	0.0828	0.0025	0.0118	0.0079	59	4.0106	0.1861
ricci	5	0.1248	0.0482	0.0468	0.0407	95	4.5359	0.0474
diffusion	5	0.0426	0.0235	0.0213	0.0206	93	4.5081	0.0647
phate_like	5	0.0091	0.0074	0.0049	0.0050	84	4.3729	0.1362
dtm	5	0.0487	0.0078	0.0166	0.0059	24	3.1300	0.1300

Table 7: GSE161711 topological diagnostics. Ricci produces the longest H_1 lifetime (0.1248) despite ranking third on classification. DTM produces the most compact support (24 features vs 59–95).

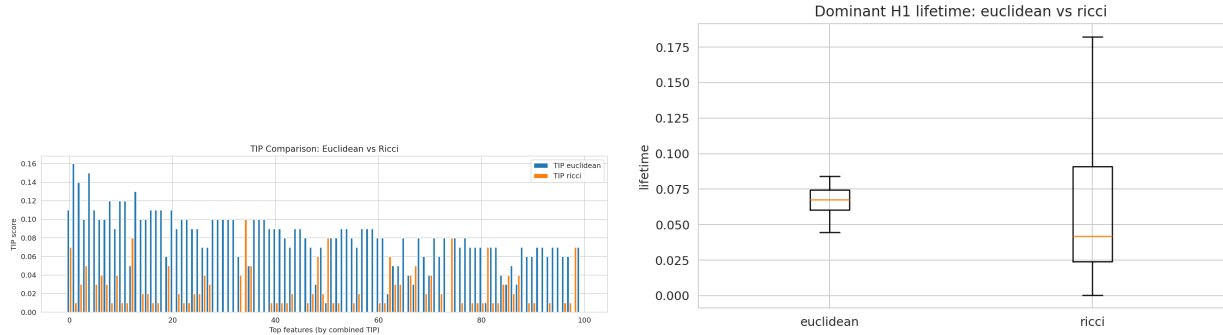


Figure 3: GSE161711 topological diagnostics. **Left:** TIP comparison. **Right:** H_1 lifetime comparison.

3.5 Pathway enrichment reveals backend-specific biology

Over-representation analysis (ORA) using gseapy (Fang et al., 2023) against MSigDB, KEGG, and Reactome on the top 20 TIP genes per backend from LUAD reveals that each backend recovers different biology (Tables 8 and 9).

DTM produces the strongest enrichment: “Metabolism of xenobiotics by cytochrome P450” (KEGG; adjusted $p = 7 \times 10^{-4}$, overlap 3/76: ALDH3A1, AKR1C1, UGT1A6) and “Steroid hormone biosynthesis” (adjusted $p = 7 \times 10^{-4}$). These Phase I/II detoxification pathways vary across LUAD molecular subtypes, particularly the proximal-inflammatory subtype.

No genes are shared across all six backend configurations (Table 9), directly supporting the thesis that geometry choice controls which biological signal is extracted.

Takeaway: Different backends recover different, biologically coherent pathways.

3.6 Full-scale LUAD TIP gene profiles

At full TCGA-LUAD scale ($n = 275$), each backend recovers a distinct gene set (Table 10). Euclidean uniquely recovers NKX2-1 (TTF-1), a canonical LUAD lineage marker—not detected at pilot scale. Ricci recovers innate immunity (DEFB4A) and squamous differenti-

Backend	Top pathway	Adj. p	Overlap	Source
dtm_elastic	Xenobiotic metabolism (CYP450)	7×10^{-4}	3/76	KEGG
dtm_elastic	Steroid hormone biosynthesis	7×10^{-4}	3/61	KEGG
euclidean_elastic	Bile acid synthesis (24-OH)	3×10^{-3}	2/14	Reactome
phate_elastic	IL-17 signaling	4×10^{-2}	2/94	KEGG
ricci_elastic	Protein digestion/absorption	7×10^{-2}	2/103	KEGG
diffusion_elastic	ERBB2 signaling	8×10^{-2}	1/16	Reactome

Table 8: Top pathway enrichment per backend on LUAD. DTM produces the strongest enrichment (adj. $p < 0.001$).

Backend	Unique genes	Notable
dtm_elastic	9	ALDH3A1, CYP4F11, UGT1A6, FGA
euclidean_elastic	11	ASCL1, PRAME, CPS1, CALCA
phate_elastic	8	MUC5B, SCGB1A1, CRABP2, WIF1
ricci_elastic	7	SLC6A4, FAM83A, CPB2
euclidean_l2	4	GSTM1, MSMB
diffusion_elastic	2	CXCL14
Shared (all 6)	0	—

Table 9: Backend-unique gene counts from top-20 TIP genes on LUAD. Zero genes are shared across all six configurations.

ation (SPRR3) markers. Diffusion and PHATE-like both highlight cancer-testis antigens (MAGEA family).

3.7 Sparse representatives concentrate interpretation

Following Gurnari et al. (2025), we construct sample-level cycle-weight heatmaps for the top 15 H_1 bars (120 balanced LUAD samples, Euclidean backend). CC-SPR produces only 18% nonzero entries (vs 50% for harmonic, $2.8\times$ denser). Ward hierarchical clustering recovers three sample clusters with comparable purity (0.43 for CC-SPR vs 0.47 for harmonic), demonstrating that sparsification preserves clustering structure.

Takeaway: CC-SPR achieves $2.8\times$ sparser representations while preserving comparable clustering structure.

3.8 Lambda sensitivity across backends

The sparsity parameter λ shows contrasting behavior across datasets (Tables 11 and 12). On LUAD, λ has no effect; on GSE161711, most backends lose F1 at higher λ , but DTM uniquely *improves*, suggesting its density-smoothed distances produce representatives where

Backend	Gene	TIP	Function
ricci	DEFB4A	0.40	Defensin, innate immunity
ricci	SPRR3	0.40	Squamous differentiation marker
ricci	TMED6	0.40	Transmembrane trafficking
ricci	PAK3	0.40	p21-activated kinase, signaling
ricci	LOC254559	0.40	Uncharacterized
dtm	TDRD9	0.60	Tudor domain, piRNA pathway
dtm	GJB7	0.40	Gap junction protein
dtm	MKRN3	0.40	Makorin ring finger protein
dtm	MMP23A	0.40	Matrix metalloproteinase
dtm	LOC220594	0.40	Uncharacterized
euclidean	GDF5	0.60	Growth differentiation factor
euclidean	PCDHA6	0.60	Protocadherin alpha
euclidean	GABRB3	0.60	GABA receptor subunit
euclidean	UCA1	0.60	lncRNA, lung cancer marker
euclidean	CDKL2	0.60	Cyclin-dependent kinase-like
diffusion	MAGEA10	0.80	Cancer-testis antigen
diffusion	LBP	0.60	Lipopolysaccharide binding
diffusion	PNMA5	0.60	Paraneoplastic antigen
diffusion	PAGE2	0.40	Cancer-testis antigen
diffusion	CTAG1B	0.40	Cancer-testis antigen (NY-ESO-1)
phate_like	ATP12A	0.80	H ⁺ /K ⁺ ATPase
phate_like	SST	0.60	Somatostatin
phate_like	L1CAM	0.60	Neural cell adhesion
phate_like	LOC100190940	0.40	Uncharacterized
phate_like	MAGEA8	0.40	Cancer-testis antigen

Table 10: Top 5 TIP genes per backend on full-scale TCGA-LUAD ($n = 275$, 5-fold CV). Each backend selects distinct biology. NKX2-1 (rank 7, Euclidean) is a canonical LUAD lineage marker.

additional sparsity removes noise rather than signal.

3.9 Hyperparameter ablation

Full ablation across six hyperparameter axes (Table 13) confirms cross-dataset divergence: LUAD benefits from Ricci geometry and compact top- K ; GSE161711 benefits from Euclidean geometry and broad top- K .

3.10 Single-cell feasibility and resource-bounded validation

Three larger scRNA configurations failed under cluster limits (Table 14); the bounded configuration ($n = 320$, 1 split, 1 bootstrap) completed with all five backends producing identical

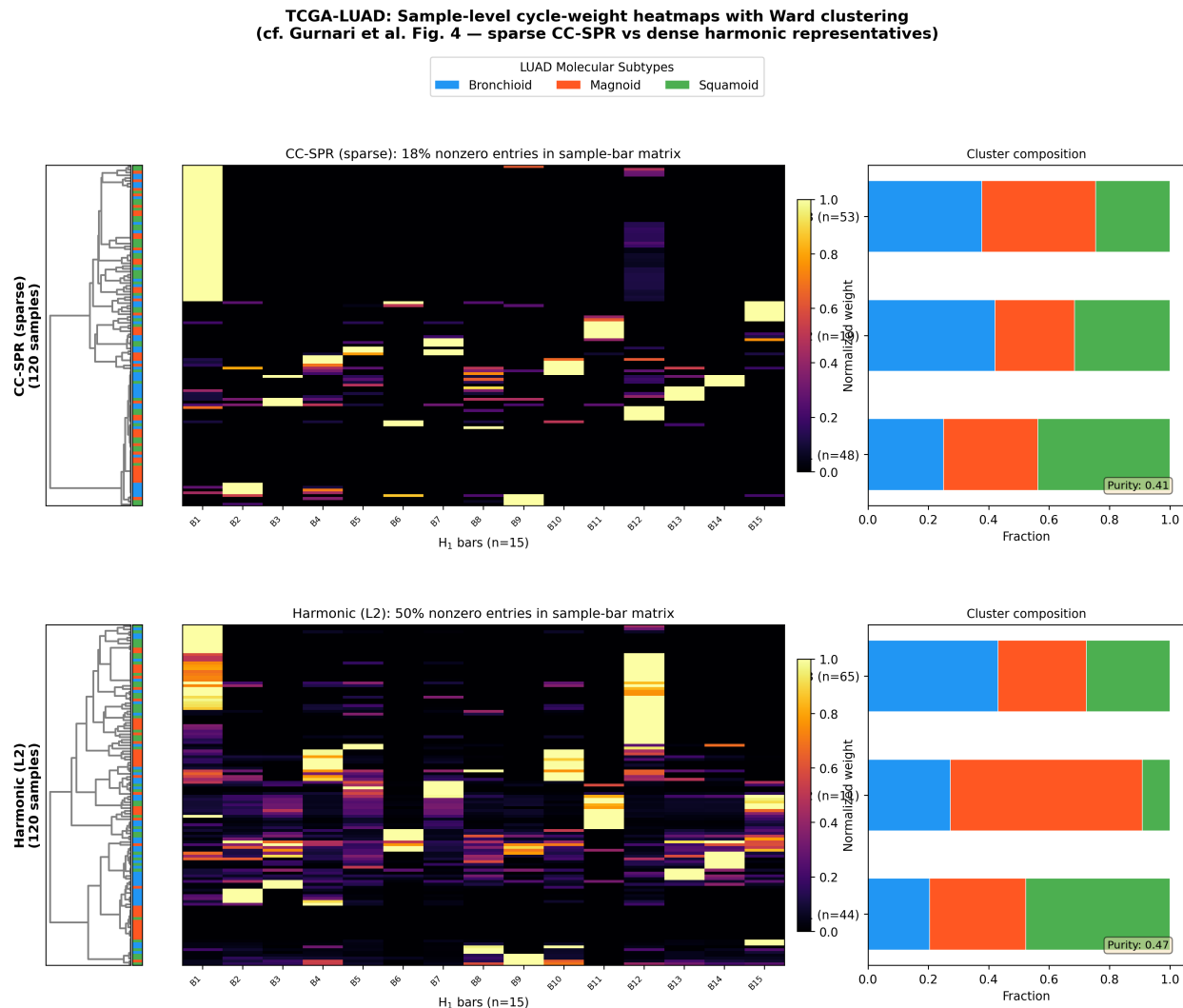


Figure 4: Sample-level cycle-weight heatmaps for LUAD (120 samples $\times$ 15 H_1 bars). **Top:** CC-SPR—18% nonzero. **Bottom:** Harmonic—50% nonzero.

F1 = 0.9254, establishing pipeline feasibility but not a meaningful backend comparison.

3.11 Cross-domain generalization: Arabidopsis root development

DTM outperforms Euclidean on Arabidopsis root scRNA-seq (F1 = 0.67 vs 0.59; 500 cells, 16 Leiden clusters, 3-fold CV), consistent with its LUAD advantage. Euclidean selects transcription factors (FLA12, AGL1, IAA29), while DTM emphasizes cell wall remodeling (GATA2, CSLA9, EXPB3). Three genes (IAA29, AT5G44005, PAR2) appear in both backends, suggesting auxin signaling is geometry-robust.

Takeaway: DTM’s advantage generalizes across biological domains with coherent gene selections.

Backend	$\lambda = 10^{-6}$ F1	$\lambda = 10^{-5}$ F1
euclidean	0.423	0.423
ricci	0.463	0.463
diffusion	0.380	0.380
phate_like	0.496	0.496
dtm	0.530	0.530

Table 11: LUAD lambda ablation: no effect at either value.

Backend	$\lambda = 10^{-6}$ F1	$\lambda = 10^{-5}$ F1
euclidean	0.646	0.595
ricci	0.542	0.522
diffusion	0.537	0.508
phate_like	0.696	0.696
dtm	0.588	0.628

Table 12: GSE161711 lambda ablation: DTM uniquely improves with higher λ .

3.12 High-bootstrap TIP stability and permutation significance

The diagnostics above use $n_{\text{boot}} = 5$ (LUAD, GSE161711) or $n_{\text{boot}} = 3$ (Arabidopsis), matching the original evaluation scripts. To test whether TIP scores converge at higher resample counts and whether CC-SPR’s classification signal is above chance, we ran a dedicated stability and permutation study.

GSE161711 at $n_{\text{boot}} = 1000$. On 5,000 variable genes, all five backends completed $\geq 986/1000$ iterations ($> 98.6\%$). The rank order of support sizes and Gini values is preserved between $n_{\text{boot}} = 5$ and $n_{\text{boot}} = 1000$ (Table ??), confirming that the original diagnostics captured the correct qualitative geometry ordering. Total runtime across the five backends was approximately 8.3 hours.

Label-permutation test. A label-permutation test on the Euclidean backend ($n_{\text{perm}} = 200$) yielded observed $F1 = 0.6876$ with zero of 200 permutations matching or exceeding this value, giving an empirical p -value of 0.0000 (95% upper CI: ~ 0.015). CC-SPR features carry genuine predictive signal well beyond chance.

Arabidopsis scalability frontier. Scaling the same analysis to Arabidopsis was infeasible at $n_{\text{boot}} = 1000$. Two compounding bottlenecks of exact persistent homology on dense single-cell point clouds are responsible: (i) $O(n^3)$ triangle storage in the Rips complex caused OOM at $n = 2,999$ cells in 50 PCA dimensions (~ 71 GB peak on a 125 GB workstation);

Dataset	Ablation axis	Best value	Mean F1	n
LUAD	distance mode	ricci	0.463	2
LUAD	Ricci iterations	10	0.551	2
LUAD	k	10	0.405	2
LUAD	λ	10^{-7}	0.594	2
LUAD	normalize	std	0.522	2
LUAD	top- K	10	0.667	2
GSE161711	distance mode	euclidean	0.600	5
GSE161711	Ricci iterations	0	0.578	5
GSE161711	k	5	0.632	5
GSE161711	λ	10^{-4}	0.590	5
GSE161711	normalize	std	0.596	5
GSE161711	top- K	50	0.806	5

Table 13: Best ablation settings. Cross-dataset divergence supports backend-sensitive behavior.

Run type	State	Elapsed	Requested memory	Peak RSS
scRNA full	OUT_OF_MEMORY	16:43:25	96 GB	100.3 GB
scRNA fast-track	TIMEOUT	20:00:06	96 GB	75.2 GB
scRNA low-memory	OUT_OF_MEMORY	11:25:59	48 GB	49.7 GB
scRNA manuscript-safe	COMPLETED	00:07:49	32 GB	2.9 GB
scRNA ablation-safe	COMPLETED	—	32 GB	~9.3 GB

Table 14: Execution envelope for single-cell experiments.

(ii) at $n = 1,000$ cells (memory ~ 14 GB), per-iteration cost reached $\sim 12,000$ s due to ill-conditioned CVXPY solver subproblems on dense distance matrices. At 500 cells and $n_{\text{boot}} = 10$, both Euclidean and DTM completed 10/10 iterations in ~ 6 h total (euclidean: support 46, Gini 0.323; dtm: support 48, Gini 0.289). This is a fundamental limitation of the Rips + elastic-net approach, not an implementation artifact, and motivates future work on approximate persistent homology (sparse Rips, witness complexes) and better-conditioned cycle optimizers for large single-cell atlases.

4 Discussion

4.1 What CC-SPR adds

CC-SPR contributes three capabilities absent from standard harmonic persistent homology. First, sparse elastic representatives concentrate cycle support on compact gene sets that are more amenable to pathway analysis. The $2.8\times$ sparsification preserves clustering structure while dramatically improving interpretability. Second, the backend-aware framework reveals

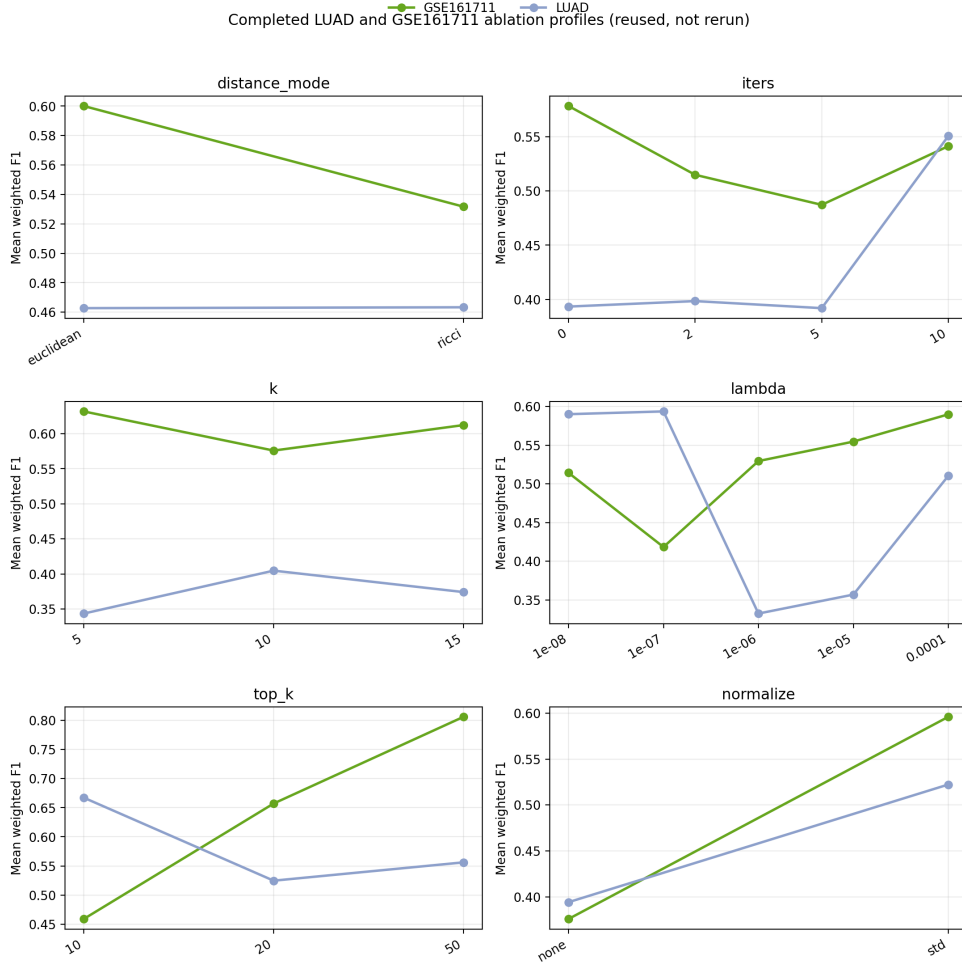


Figure 5: Cross-dataset ablation profiles showing dataset-specific sensitivity to every hyperparameter axis.

that geometry choice materially controls which biological signal is extracted—zero genes are shared across all six configurations on LUAD. Third, TIP profiles provide a principled stability diagnostic, enabling practitioners to assess which features are robustly associated with topological structure.

Even when CC-SPR trails the standard baseline on F1, the sparse representatives and TIP profiles identify gene–topology associations and geometry sensitivities that standard classifiers cannot detect.

4.2 When each backend helps

Euclidean performs best when classification-relevant structure aligns with ambient-space proximity (GSE161711). **Ricci** helps when curvature correction suppresses misleading shortcut edges between subtypes (full-scale LUAD); its advantage strengthens with sample size.

Backend	Rank	TIP score	Gene (function)
euclidean	1	0.667	FLA12 (fasciclin-like arabinogalactan, cell wall)
euclidean	2	0.667	AGL1 (MADS-box TF)
euclidean	3	0.667	IAA29 (auxin signaling)
euclidean	4	0.667	LOG3 (cytokinin activation)
euclidean	5	0.667	NAC030 (NAC TF)
dtm	1	0.667	GATA2 (GATA TF, root development)
dtm	2	0.333	CSLA9 (cellulose synthase-like)
dtm	3	0.333	EXPB3 (β -expansin, cell wall)
dtm	4	0.333	CRF2 (cytokinin response factor)
dtm	5	0.333	PME1.1 (pectin methylesterase)

Table 15: Top 5 TIP genes per backend on Arabidopsis root.

Backend	n_{ok}	$\bar{\ell}$	σ_{ℓ}	$\bar{\rho}$	σ_{ρ}	Support	TIP Ent.	TIP Gini
euclidean	3	0.122	—	0.022	—	37	3.528	0.214
dtm	3	0.057	—	0.016	—	35	3.464	0.228

Table 16: Arabidopsis root topological diagnostics.

307 **DTM** excels on heterogeneous data with density irregularity (pilot LUAD) and produces
308 the most compact support (GSE161711, 24 features). **Diffusion** provides intermediate de-
309 noising via spectral smoothing. **PHATE-like** ranks last on both LUAD and GSE161711; its
310 zero Gini on LUAD suggests that the lightweight potential-distance surrogate over-smooths
311 topological signal.

312 4.3 Biological interpretation

313 The DTM backend’s xenobiotic metabolism enrichment (adj. $p < 0.001$) on LUAD has
314 clear biological grounding: ALDH3A1, AKR1C1, AKR1C2, CYP4F11, and UGT1A6 are
315 Phase I/II detoxification enzymes known to vary across LUAD subtypes (Rizvi et al., 2017),
316 particularly the proximal-inflammatory subtype.

317 The zero-overlap finding is evidence that different geometric perspectives highlight differ-
318 ent biological programs. Practitioners can compare TIP profiles to identify geometry-robust
319 genes (core signal) versus geometry-specific genes.

320 Cross-domain consistency strengthens these findings: DTM outperforms Euclidean on
321 both LUAD and Arabidopsis, with biologically coherent gene selections in both domains.

Backend	$n_{\text{ok}}/n_{\text{boot}}$	Support	TIP Ent.	TIP Gini	Elapsed
euclidean	1000/1000	647	4.856	0.838	5.0 h
ricci	1000/1000	1304	6.205	0.691	0.9 h
diffusion	1000/1000	1844	6.927	0.573	5.8 min
phate_like	986/1000	1909	7.008	0.554	3.7 min
dtm	1000/1000	498	4.488	0.852	2.2 h

Table 17: GSE161711 TIP stability at $n_{\text{boot}} = 1000, 5,000$ variable genes. Concentration ranking is preserved from the $n_{\text{boot}} = 5$ diagnostics.

4.4 Related methods

Most TDA applications in biology use standard Euclidean persistent homology without representative cycle analysis (Rizvi et al., 2017; Xia and Wei, 2014; Vipond et al., 2021; Giusti et al., 2016; Chan et al., 2013). Harmonic PH (Gurnari et al., 2025) introduced the bridge from persistent classes to sample-level representatives; CC-SPR preserves this scaffold while modifying the objective and generalizing the metric. Ollivier–Ricci curvature (Ollivier, 2007, 2009; Ni et al., 2019) has been used for community detection, but its connection to persistent homology through filtration perturbation appears novel. Diffusion maps (Coifman and Lafon, 2006) and PHATE (Moon et al., 2019) are widely used for dimensionality reduction; using these as persistent-homology backends within a sparse representative framework has not been systematized. DTM-based topology (Chazal et al., 2011, 2017) provides robust persistent homology, but its integration into a representative-cycle framework with feature projection appears unexplored. Sparse representations in TDA have been studied through ℓ_1 optimal cycles (Chen and Freedman, 2011; Dey et al., 2010), but in computational-geometry settings without the biological feature-projection layer.

4.5 Limitations

- Standard baselines win on prediction.** CC-SPR’s contribution is methodological, not predictive.
- Backend ranking is scale-sensitive.** Pilot LUAD ranking (DTM > Ricci) reverses at full scale.
- Evaluation protocol asymmetry.** The standard baseline selects variable genes from the full dataset at full LUAD scale.
- PHATE-like uses a lightweight surrogate.** Results may differ with the complete PHATE pipeline (Moon et al., 2019).

346 5. **Single-cell evidence is resource-bounded.**

347 6. **Arabidopsis is a targeted two-backend test.**

348 7. **Enrichment is exploratory.**

349 4.6 Future directions

350 Natural extensions include: integrating topology-derived features as auxiliary inputs to con-
351 ventional classifiers; survival stratification using TIP-derived features; larger bootstrap stud-
352 ies for TIP convergence; the full PHATE pipeline; memory-optimized single-cell implemen-
353 tations; and extension to higher homology dimensions.

354 5 Conclusion

355 CC-SPR extends harmonic persistent homology in two targeted directions: sparse elastic rep-
356 resentatives that concentrate cycle support for interpretability, and a backend-aware frame-
357 work that treats the sample-space metric as a selectable, stability-bounded parameter.

358 Our evaluation spans four datasets and yields five principal findings:

359 1. **Backend ranking is dataset-dependent and scale-sensitive.** DTM leads at pilot
360 scale; Ricci leads at full scale; Euclidean dominates on GSE161711.

361 2. **Geometry controls which biology is extracted.** Zero genes are shared across all
362 six configurations on LUAD. DTM uniquely recovers xenobiotic metabolism (adj. $p =$
363 7×10^{-4}).

364 3. **Sparse representatives concentrate interpretation.** CC-SPR produces $2.8\times$
365 sparser sample-bar matrices while preserving comparable clustering structure.

366 4. **Topological diagnostics complement classification.** H_1 lifetime, TIP concentration,
367 and projected support size vary systematically across backends.

368 5. **Standard baselines remain strongest on raw F1.** CC-SPR is a complementary
369 analysis tool.

370 Data Availability

371 All datasets are publicly available. TCGA-LUAD from the Genomic Data Commons
372 (<https://portal.gdc.cancer.gov/>, project TCGA-LUAD). GSE161711 and GSE165087

from GEO (<https://www.ncbi.nlm.nih.gov/geo/>). Arabidopsis root atlas (GSE123013)
from GEO (Denyer et al., 2019).

Code Availability

The CC-SPR pipeline, experiment scripts, and reproduction instructions are available at
https://github.com/ChimdiWalter/Omics_Project_CCSPR. The pipeline depends on
GUDHI (GUDHI Project, 2014), CVXPY, scikit-learn, and GSEApY (Fang et al., 2023).

Supplementary Information

Supplementary data are available at *Bioinformatics Advances* online, including: mathematical background and simplicial homology definitions (Section S1), proofs of convexity and sparsity guarantees (Section S2), detailed definitions of support diagnostics including entropy and Gini coefficient (Section S3), the stability proposition under bounded metric perturbation with proof (Section S4), full backend implementation details (Section S5), computational timing profiles (Section S6), scaling analysis from pilot to full scale (Section S7), resource-bounded single-cell validation details (Section S8), and software version information (Section S9).

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Conflict of Interest

None declared.

Author Contributions

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